msk_top_regs address map

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0xAC

MSK Modem Configuration and Status Registers

Offset	Identifier	Name
0x00	Hash_ID_Low	Pluto MSK FPGA Hash ID - Lower 32-bits
0x04	Hash_ID_High	Pluto MSK FPGA Hash ID - Upper 32-bits
0x08	MSK_Init	MSK Modem Initialization Control
0x0C	MSK_Control	MSK Modem Control
0x10	MSK_Status	MSK Modem Status 0
0x14	Tx_Bit_Count	MSK Modem Status 1
0x18	Tx_Enable_Count	MSK Modem Status 2
0x1C	Fb_FreqWord	Bitrate NCO Frequency Control Word
0x20	TX_F1_FreqWord	Tx F1 NCO Frequency Control Word
0x24	TX_F2_FreqWord	Tx F2 NCO Frequency Control Word
0x28	RX_F1_FreqWord	Rx F1 NCO Frequency Control Word
0x2C	RX_F2_FreqWord	Rx F2 NCO Frequency Control Word
0x30	LPF_Config_0	PI Controller Configuration and Low-pass Filter Configuration
0x34	LPF_Config_1	PI Controller Configuration Configuration Register 1
0x38	Tx_Data_Width	Modem Tx Input Data Width
0x3C	Rx_Data_Width	Modem Rx Output Data Width
0x40	PRBS_Control	PRBS Control 0
0x44	PRBS_Initial_State	PRBS Control 1
0x48	PRBS_Polynomial	PRBS Control 2
0x4C	PRBS_Error_Mask	PRBS Control 3
0x50	PRBS_Bit_Count	PRBS Status 0
0x54	PRBS_Error_Count	PRBS Status 1
0x58	LPF_Accum_F1	F1 PI Controller Accumulator
0x5C	LPF_Accum_F2	F2 PI Controller Accumulator
0x60	axis_xfer_count	MSK Modem Status 3
0x64	Rx_Sample_Discard	Rx Sample Discard
0x68	LPF_Config_2	PI Controller Configuration Configuration Register 2
0x6C	f1_nco_adjust	F1 NCO Frequency Adjust
0x70	f2_nco_adjust	F2 NCO Frequency Adjust
0x74	f1_error	F1 Error Value
0x78	f2_error	F2 Error Value
0x7C	Tx_Sync_Ctrl	Transmitter Sync Control
0x80	Tx_Sync_Cnt	Transmitter Sync Duration
0x84	Tx_Sync_Pat	Transmitter Sync pattern

Offset	Identifier	Name
0x88	lowpass_ema_alpha1	Exponential Moving Average Alpha
0x8C	lowpass_ema_alpha2	Exponential Moving Average Alpha
0x90	rx_power	Receive Power
0x94	tx_async_fifo_rd_wr_ptr	Tx async FIFO read and write pointers
0x98	rx_async_fifo_rd_wr_ptr	Rx async FIFO read and write pointers
0x9C	rx_frame_sync_status	Frame Sync Status
0xA0	symbol_lock_control	Symbol Lock Control
0xA4	symbol_lock_status	Symbol Lock Status
0xA8	symbol_lock_time	Symbol Lock Time

Hash_ID_Low register

- Absolute Address: 0x0
- Base Offset: 0x0
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	hash_id_lo	r	0xAAAA5555	Hash ID Lower 32-bits

hash_id_lo field Lower 32-bits of Pluto MSK FPGA Hash ID

Hash_ID_High register

- Absolute Address: 0x4
- Base Offset: 0x4
- Size: 0x4

Bits	Identifier	Access	Reset	Name
31:0	hash_id_hi	r	0x5555AAAA	Hash ID Upper 32-bits

hash_id_hi field Upper 32-bits of Pluto MSK FPGA Hash ID

MSK_Init register

- Absolute Address: 0x8
- Base Offset: 0x8
- Size: 0x4

Synchronous initialization of MSK Modem functions, does not affect configuration registers.

Bits	Identifier	Access	Reset	Name
0	txrxinit	rw	0x1	Tx/Rx Init Enable
1	txinit	rw	0x1	Tx Init Enable
2	rxinit	rw	0x1	Rx Init Enable

txrxinit field 0 -> Normal modem operation

1 -> Initialize Tx and Rx

txinit field 0 -> Normal Tx operation

1 -> Initialize Tx

rxinit field 0 -> Normal Rx operation

1 -> Initialize Rx

MSK_Control register

- Absolute Address: 0xC
- Base Offset: 0xC
- Size: 0x4

MSK Modem Configuration and Control

Bits	Identifier	Access	Reset	Name
0	ptt	rw	0x0	Push-to-Talk Enable
1	loopback_ena	rw	0x0	Modem Digital Tx -> Rx Loopback Enable
2	rx_invert	rw	0x0	Rx Data Invert Enable
3	clear_counts	rw	0x0	Clear Status Counters
4	diff_encoder_loopback	rw	0x0	Differential Encoder -> Decoder Loopback Enable
7:5	reserved	rw	0x0	—
10:8	tx_shift	rw	0x0	Tx Sample Shoift

ptt field 0 -> PTT Disabled

1 -> PTT Enabled

loopback_ena field 0 -> Modem loopback disabled

1 -> Modem loopback enabled

rx_invert field 0 -> Rx data normal

1 -> Rx data inverted

clear_counts field Clear Tx Bit Counter and Tx Enable Counter

diff_encoder_loopback field 0 -> Differential Encoder -> Decoder loopback disabled

1 -> Differential Encoder -> Decoder loopback enabled

reserved field Reserved

tx_shift field Shift left from 0 to 4 bits

MSK_Status register

- Absolute Address: 0x10
- Base Offset: 0x10
- Size: 0x4

Modem status bits

Bits	Identifier	Access	Reset	Name
0	demod_sync_lock	r	0x0	Demodulator Sync Status
1	tx_enable	r	0x0	AD9363 DAC Interface Tx Enable Input Active
2	rx_enable	r	0x0	AD9363 ADC Interface Rx Enable Input Active
3	tx_axis_valid	r	0x0	Tx S_AXIS_VALID

demod_sync_lock field Demodulator Sync Status - not currently implemented

tx_enable field 1 -> Data to DAC Enabled

0 -> Data to DAC Disabled

rx_enable field 1 -> Data from ADC Enabled

0 -> Data from ADC Disabled

tx_axis_valid field 1 -> S_AXIS_VALID Enabled

0 -> S_AXIS_VALID Disabled

Tx_Bit_Count register

- Absolute Address: 0x14
- Base Offset: 0x14
- Size: 0x4

Modem status data

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	Tx Bit Count

data field Count of data requests made by modem

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

Tx_Enable_Count register

- Absolute Address: 0x18
- Base Offset: 0x18
- Size: 0x4

Modem status data

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	Tx Enable Count

data field Number of clocks on which Tx Enable is active

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

Fb_FreqWord register

- Absolute Address: 0x1C
- Base Offset: 0x1C
- Size: 0x4

Set Modem Data Rate

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	Frequency Control Word

config_data field Sets the center frequency of the NCO as FW = Fn * 2^32/Fs, where Fn is the desired NCO frequency, and Fs is the NCO sample rate

TX_F1_FreqWord register

- Absolute Address: 0x20
- Base Offset: 0x20
- Size: 0x4

Set Modulator F1 Frequency

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	Frequency Control Word

config_data field Sets the center frequency of the NCO as FW = Fn * 2^32/Fs, where Fn is the desired NCO frequency, and Fs is the NCO sample rate

TX_F2_FreqWord register

- Absolute Address: 0x24
- Base Offset: 0x24
- Size: 0x4

Set Modulator F2 Frequency

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	Frequency Control Word

config_data field Sets the center frequency of the NCO as FW = Fn * 2^32/Fs, where Fn is the desired NCO frequency, and Fs is the NCO sample rate

RX_F1_FreqWord register

- Absolute Address: 0x28
- Base Offset: 0x28
- Size: 0x4

Set Demodulator F1 Frequency

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	Frequency Control Word

config_data field Sets the center frequency of the NCO as FW = Fn * 2^32/Fs, where Fn is the desired NCO frequency, and Fs is the NCO sample rate

RX_F2_FreqWord register

- Absolute Address: 0x2C
- Base Offset: 0x2C
- Size: 0x4

Set Demodulator F2 Frequency

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	Frequency Control Word

config_data field Sets the center frequency of the NCO as FW = Fn * 2^32/Fs, where Fn is the desired NCO frequency, and Fs is the NCO sample rate

LPF_Config_0 register

- Absolute Address: 0x30
- Base Offset: 0x30
- Size: 0x4

Configure PI controller and low-pass filter

Bits	Identifier	Access	Reset	Name
0	lpf_freeze	rw	0x0	Freeze the accumulator's current value
1	lpf_zero	rw	0x0	Hold the PI Accumulator at zero
7:2	prbs_reserved	rw	0x0	Reserved
31:8	lpf_alpha	rw	0x0	Lowpass IIR filter alpha

lpf_freeze field 0 -> Normal operation

1 -> Freeze current value

lpf_zero field 0 -> Normal operation

1 -> Zero and hold accumulator

lpf_alpha field Value controls the filter rolloff

LPF_Config_1 register

- Absolute Address: 0x34
- Base Offset: 0x34
- Size: 0x4

Configures PI Controller I-gain and divisor

Bits	Identifier	Access	Reset	Name
23:0	i_gain	rw	0x0	Integral Gain Value
31:24	i_shift	rw	0x0	Integral Gain Bit Shift

i_gain field Value m of 0-16,777,215 sets the integral multiplier

i_shift field Value n of 0-32 sets the integral divisor as 2^-n

Tx_Data_Width register

- Absolute Address: 0x38
- Base Offset: 0x38
- Size: 0x4

Set the parallel data width of the parallel-to-serial converter

Bits	Identifier	Access	Reset	Name
7:0	data_width	rw	0x8	Modem input/output data width

data_width field Set the data width of the modem input/output

Rx_Data_Width register

- Absolute Address: 0x3C
- Base Offset: 0x3C
- Size: 0x4

Set the parallel data width of the serial-to-parallel converter

Bits	Identifier	Access	Reset	Name
7:0	data_width	rw	0x8	Modem input/output data width

data_width field Set the data width of the modem input/output

PRBS_Control register

- Absolute Address: 0x40
- Base Offset: 0x40
- Size: 0x4

Configures operation of the PRBS Generator and Monitor

Bits	Identifier	Access	Reset	Name
0	prbs_sel	rw	0x0	PRBS Data Select
1	prbs_error_insert	w	0x0	PRBS Error Insert
2	prbs_clear	w	0x0	PRBS Clear Counters
3	prbs_manual_sync	w	0x0	PRBS Manual Sync
15:4	prbs_reserved	rw	0x0	Reserved
31:16	prbs_sync_threshold	rw	0x0	PRBS Auto Sync Threshold

prbs_sel field 0 -> Select Normal Tx Data 1 -> Select PRBS Tx Data

prbs_error_insert field 0 -> 1 : Insert bit error in Tx data (both Normal and PRBS)

1 -> 0 : Insert bit error in Tx data (both Normal and PRBS)

prbs_clear field 0 -> 1 : Clear PRBS Counters

1 -> 0 : Clear PRBS Counters

prbs_manual_sync field 0 -> 1 : Synchronize PRBS monitor

1 -> 0 : Synchronize PRBS monitor

prbs_sync_threshold field 0 : Auto Sync Disabled

N > 0 : Auto sync after N errors

PRBS_Initial_State register

- Absolute Address: 0x44
- Base Offset: 0x44
- Size: 0x4

PRBS Initial State

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	PRBS Seed

config_data field Sets the starting value of the PRBS generator

PRBS_Polynomial register

- Absolute Address: 0x48
- Base Offset: 0x48
- Size: 0x4

PRBS Polynomial

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	PRBS Polynomial

config_data field Bit positions set to '1' indicate polynomial feedback positions

PRBS_Error_Mask register

- Absolute Address: 0x4C
- Base Offset: 0x4C
- Size: 0x4

PRBS Error Mask

Bits	Identifier	Access	Reset	Name
31:0	config_data	rw	0x0	PRBS Error Mask

config_data field Bit positions set to '1' indicate bits that are inverted when a bit error is inserted

PRBS_Bit_Count register

- Absolute Address: 0x50
- Base Offset: 0x50
- Size: 0x4

PRBS Bits Received

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	PRBS Bits Received

data field Number of bits received by the PRBS monitor since last BER can be calculated as the ratio of received bits to errored-bits

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

PRBS_Error_Count register

- Absolute Address: 0x54
- Base Offset: 0x54
- Size: 0x4

PRBS Bit Errors

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	PRBS Bit Errors

data field Number of errored-bits received by the PRBS monitor since last sync BER can be calculated as the ratio of received bits to errored-bits

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

LPF_Accum_F1 register

- Absolute Address: 0x58
- Base Offset: 0x58
- Size: 0x4

Value of the F1 PI Controller Accumulator

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	PI Controller Accumulator Value

data field PI Controller Accumulator Value

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

LPF_Accum_F2 register

- Absolute Address: 0x5C
- Base Offset: 0x5C
- Size: 0x4

Value of the F2 PI Controller Accumulator

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	PI Controller Accumulator Value

data field PI Controller Accumulator Value

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

axis_xfer_count register

- Absolute Address: 0x60
- Base Offset: 0x60
- Size: 0x4

Modem status data

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	S_AXIS Transfers

data field Number completed S_AXIS transfers

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

Rx_Sample_Discard register

- Absolute Address: 0x64
- Base Offset: 0x64
- Size: 0x4

Configure samples discard operation for demodulator

Bits	Identifier	Access	Reset	Name
7:0	rx_sample_discard	rw	0x0	Rx Sample Discard Value
15:8	rx_nco_discard	rw	0x0	Rx NCO Sample Discard Value

rx_sample_discard field Number of Rx samples to discard

rx_nco_discard field Number of NCO samples to discard

LPF_Config_2 register

- Absolute Address: 0x68
- Base Offset: 0x68
- Size: 0x4

Configures PI Controller I-gain and divisor

Bits	Identifier	Access	Reset	Name
23:0	p_gain	rw	0x0	Proportional Gain Value
31:24	p_shift	rw	0x0	Proportional Gain Bit Shift

p_gain field Value m of 0-16,777,215 sets the proportional multiplier

p_shift field Value n of 0-32 sets the proportional divisor as 2ˆ-n

f1_nco_adjust register

- Absolute Address: 0x6C
- Base Offset: 0x6C
- Size: 0x4

Status Register

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Frequency offet applied to the F1 NCO

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

f2_nco_adjust register

- Absolute Address: 0x70
- Base Offset: 0x70
- Size: 0x4

Status Register

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Frequency offet applied to the F2 NCO

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

f1_error register

- Absolute Address: 0x74
- Base Offset: 0x74
- Size: 0x4

Status Register

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Error value of the F1 Costas loop after each active bit period

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

f2_error register

- Absolute Address: 0x78
- Base Offset: 0x78
- Size: 0x4

Status Register

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Error value of the F2 Costas loop after each active bit period

This register is write-to-capture.

To read data the following steps are required:

1 - Write any value to this register to capture read data

2 - Read the register

Tx_Sync_Ctrl register

- Absolute Address: 0x7C
- Base Offset: 0x7C
- Size: 0x4

Provides control bits for generation of transmitter synchronization patterns

Bits	Identifier	Access	Reset	Name
0	tx_sync_ena	rw	0x0	Tx Sync Enable
1	tx_sync_force	rw	0x0	Tx Sync Force

tx_sync_ena field 0 : Disable sync transmission

1 : Enable sync transmission when PTT is asserted

tx_sync_force field 0 : Normal operation

1 : Continuously transmit synchronization pattern

Tx_Sync_Cnt register

- Absolute Address: 0x80
- Base Offset: 0x80
- Size: 0x4

Sets the duration of the synchronization tones when enabled

Bits	Identifier	Access	Reset	Name
23:0	tx_sync_cnt	rw	0x0	Tx sync duration

tx_sync_cnt field Value from 0x00_0000 to 0xFF_FFFF.

This value represents the number bit-times the synchronization signal should be sent after PTT is asserted.

Tx_Sync_Pat register

- Absolute Address: 0x84
- Base Offset: 0x84
- Size: 0x4

Sets the synchronization pattern to be transmitted when synchronization tones are enabled

Bits	Identifier	Access	Reset	Name
15:0	tx_sync_pat	rw	0x1B33	Tx sync pattern

tx_sync_pat field Value from 0x0000 to 0xFFFF.

This value represents the number bit-times the synchronization signal should be sent after PTT is asserted.

lowpass_ema_alpha1 register

- Absolute Address: 0x88
- Base Offset: 0x88
- Size: 0x4

Sets the alpha for the EMA

Bits	Identifier	Access	Reset	Name
17:0	alpha	rw	0x0	EMA alpha

alpha field Value from 0x0_0000 to 0x3_FFFF represent the EMA alpha

lowpass_ema_alpha2 register

- Absolute Address: 0x8C
- Base Offset: 0x8C
- Size: 0x4

Sets the alpha for the EMA

Bits	Identifier	Access	Reset	Name
17:0	alpha	rw	0x0	EMA alpha

alpha field Value from 0x0_0000 to 0x3_FFFF represent the EMA alpha

rx_power register

- Absolute Address: 0x90
- Base Offset: 0x90
- Size: 0x4

Receive power computed from I/Q samples

Bits	Identifier	Access	Reset	Name
22:0	data	rw	0x0	Receive Power

data field Value that represent the RMS power of the incoming signal (I-channel)

This register is write-to-capture. To read data the following steps are required:

Write any value to this register to capture read data

Read the register

tx_async_fifo_rd_wr_ptr register

- Absolute Address: 0x94
- Base Offset: 0x94
- Size: 0x4

Tx async FIFO read and write pointers

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Read and Write Pointers

Bits 31:16 - write pointer (12-bits) Bits 15:00 - read pointer (12-bits)

This register is write-to-capture. To read data the following steps are required:

Write any value to this register to capture read data

Read the register

rx_async_fifo_rd_wr_ptr register

- Absolute Address: 0x98
- Base Offset: 0x98
- Size: 0x4

Rx async FIFO read and write pointers

Bits	Identifier	Access	Reset	Name
31:0	data	rw	0x0	—

data field Read and Write Pointers

Bits 31:16 - write pointer (12-bits) Bits 15:00 - read pointer (12-bits)

This register is write-to-capture. To read data the following steps are required:

Write any value to this register to capture read data

Read the register

rx_frame_sync_status register

- Absolute Address: 0x9C
- Base Offset: 0x9C
- Size: 0x4

Bits	Identifier	Access	Reset	Name
0	frame_sync_locked	r	0x0	Frame Sync Lock
1	frame_buffer_overflow	r	0x0	Frame Buffer Overflow
25:2	frames_received	rw	0x0	Frames Received
31:26	frame_sync_errors	rw	0x0	Frames Sync Errors

frame_sync_locked field 0 - Frame sync not locked 1 - Frame sync locked

frame_buffer_overflow field 0 - Normal operation 1 - Buffer overflow

frames_received field Count of frames received. Value is 0x00_0000 to 0xFF_FFFF. Counter rolls over when max count is reached.

frame_sync_errors field Count of frame sync errors. Value is 0 to 63. Counter rolls over when max count is reached.

symbol_lock_control register

- Absolute Address: 0xA0
- Base Offset: 0xA0
- Size: 0x4

Bits	Identifier	Access	Reset	Name
9:0	symbol_lock_count	rw	0x80	Symbol Lock Integration Count
25:10	symbol_lock_threshold	rw	0x2710	Symbol Lock Threshold

symbol_lock_count field Sets the integration period in symbols. Value is from 0 to 1023.

symbol_lock_threshold field Sets the threshold value on which to declare symbol sync

symbol_lock_status register

- Absolute Address: 0xA4
- Base Offset: 0xA4
- Size: 0x4

Bits	Identifier	Access	Reset	Name
0	f1f2	r	0x0	Symbol Lock F1 and F2
1	f1	r	0x0	Symbol Lock F1
2	f2	r	0x0	Symbol Lock F2
3	unlock_f1	r	0x0	F1 unlocked since last read
4	unlock_f2	r	0x0	F2 unlocked since last read

f1f2 field 0 - Unlocked 1 - F1 and F2 locked

f1 field 0 - Unlocked 1 - F1 locked

f2 field 0 - Unlocked 1 - F2 locked

unlock_f1 field 0 - No unlock since last read 1 - One or mode unlocks since last read

unlock_f2 field 0 - No unlock since last read 1 - One or mode unlocks since last read

symbol_lock_time register

- Absolute Address: 0xA8
- Base Offset: 0xA8
- Size: 0x4

Bits	Identifier	Access	Reset	Name
15:0	f1	r	0x0	F1 Symbol Lock Time
31:16	f2	r	0x0	F2 Symbol Lock Time

f1 field Number of symbols for F1 lock since init released

f2 field Number of symbols for F2 lock since init released